ASSIGNMENT 5 — QUESTION 5

Person to Student Inheritance

JavaScript (ES6+) solution

```
function Person(name) {
    this.name = name;
}

Person.prototype.showName = function () {
    console.log(`Name: ${this.name}`);
};

function Student(name, branch) {
    Person.call(this, name);
    this.branch = branch;
}

Student.prototype = Object.create(Person.prototype);
Student.prototype.constructor = Student;

Student.prototype.showBranch = function () {
    console.log(`Branch: ${this.branch}`);
};

const student = new Student("Mohit", "CSE");
student.showName();
student.showBranch();

console.log(student instanceof Student); // true
console.log(student instanceof Person);  // true
```

Expected Output

```
Name: Mohit
Branch: CSE
true
true
```

Explanation

Person.call(this, name) initializes the name property on the new student. Object.create(Person.prototype) connects Student.prototype to Person.prototype, allowing the student to use methods from both prototypes.